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|----------------------------------------------------------------|-------------------|------------------------------------------------------------------------------------|
| <b>PLASMA SPRAY PROCESSORS</b><br><br><b>Works Instruction</b> | QF-QMS/02/018     |  |
|                                                                | Rev. 00           |                                                                                    |
|                                                                | Date : 01/03/2019 |                                                                                    |

Operations/Work/Job Activity covered by this assessment: Wicket Gate

**Required Personal Protective Equipment (PPE) :**



**Safety glasses must be worn at times.**



**Long and loose hair must be contained.**



**Appropriate footwear with substantial uppers must be worn.**



**Close fitting/protective clothing to cover arms and legs must be worn.**



**Respiratory protection devices required.**



**Oil free leather gloves and spats must be worn.**



**Hearing protection must be used when using this machine.**

**Loading & Unloading Check:**

1. Ensure no slip/trip is present in work pass and walk ways.
2. Ensure that work area is clean and clear of grease, oil and any flammable materials.
3. Refer Work Instructions PSP/WI/14 while handling of Wicket Gate.

**Operations Procedure:**

- 1) Refer Pre-Inspection Work Instructions PSP/WI/03 to pre-inspect the Wicket Gate.
- 2) If damaged is found on the coating surface, inform Alstom Power India Ltd (APIL) by email with photograph and get their approval.
- 3) Repair to be carried w.r.t. APIL procedure.
- 4) Pre-heat the Wicket Gate to 100°C to 150°C (Skin Temperature) with burner to remove oil & grease from the surface.
- 5) Mask with High Temperature Adhesive Tape/Metal Shim/Insulation tape to protect unwanted area from blasting and HVOF coating.
- 6) Ensure the Compressed Air is clean & free from water residues.
- 7) Check the humidity of the Blasting enclosure/room with hygrometer and ensure it is in the permissible limit of <80%.
- 8) Refer Final Blasting Work Instructions PSP/WI/04 for preparing surface. The surface preparation will be as per Sa3 specs.
- 9) Use virgin 24mesh Al2O3 grit for blasting. Maintain Air pressure to blast at 3.5bar with 6mm bore nozzle and keeping 200 ±50mm standoff distances.
- 10) If any pinholes/micro pinholes are observed convey to supervisor for approval.
- 11) Maintain 5 to 8 Ra microns surface roughness during blasting and check with roughness tester as per the protocol.
- 12) Load the Wicket Gate in the acoustic room and clamp it thoroughly on the chuck with proper packing to prevent clamping marks on the Wicket Gate Shaft.
- 13) Check the humidity of the Acoustic Enclosure/room with hygrometer and ensure it is in the

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permissible limit of <80%.

14) Start Exhaust Fans before starting the HP/HVOF Control Unit.

15) Open valve of Nitrogen and Oxygen Cylinders and set the pressure according to the parameter mentioned in the Technical Bulletin.

16) Start the JP8000 control unit with your Login ID. Check the Oxygen, Fuel, Water & Combustion Flow & pressure in the manual mode screen.

17) Set the following parameter:

|                        |                        |      |          |
|------------------------|------------------------|------|----------|
| i) Barrel              | 6"                     |      |          |
| ii) Oxygen Pressure    | 240+ <sub>50</sub> PSI | Flow | 1850SCFH |
| iii) Nitrogen Pressure | 55PSI                  | Flow | 23SCFH   |
| iv) Fuel Pressure      | 119PSI                 | Flow | 6GPH     |
| v) Spray Rate          | 75 to 80gms/min        |      |          |
| vi) Spray Distance     | 350±25mm               |      |          |

18) Pre-heat the powder SM 3652 in a SS tray and keep in an oven to 60°C to 70°C for 120mins. After the alarm pour the powder in the powder feeder hopper and maintain the temperature with heating blanket. Check with pyrometer the temperature of the heating blanket and ensure it is at 60°C to 65°C.

19) Select the powder WOKA 3652 recipe from the recipe menu.

20) Measure the powder flow rate (75 to 80 grams/min) as per parameter, and set the Powder feeder RPM.

21) Set the job speed and Robot programming speed, instructions will be given by Supervisor.

22) Closed the Acoustic Door and Wait for signal before igniting the JP 5000 Torch. Go to Development screen and press the gun ignition. Wait for the gun to ramp up till you get the desire combustion pressure as per the parameter.

23) Pre-heat the skin temperature of the job up to 45°C to 60°C, check with pyrometer.

24) Start Powder feeder and wait till it stabilize and spray the job layer by layers to get a uniform thickness all over.

25) While spraying mating surface cover the profile with metal shim and mount the sample specimen on the magnet.

26) Coat the mating surface (i.e. sides) alongwith specimen to get the desire thickness.

27) Check the dimension to get the 300+50µm thickness of the coating and 300-25µm to +75µm on mating surface.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                 |            |  |       |         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------|--|-------|---------|
| <b>Shutdown :</b> <ol style="list-style-type: none"> <li>1. Shut the Powder Feeder and wait for the feed lines to clear.</li> <li>2. Shut down the ignition flame and bring the robot to its home position.</li> <li>3. Cool the Wicket Gate and measure the Surface Roughness with Surface testing Machine and measure the value between 3 to 6µm Ra.</li> <li>4. Refer to Final Inspection Work Instruction PSP/WI/10 and get the approval of APIL Representative.</li> </ol> |                 |            |  |       |         |
| Assessor :                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Prasanna Wadkar | Signature: |  | Date: | 5.03.19 |
| Approved:                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Dhimant Doshi   | Signature: |  | Date: | 5.03.19 |